SKILL – 24:

Compilation Using Makefile:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-24$ gcc -Wall -Wextra -std=c11 main.c -o os_processor
main.c: In function ‘handle_signal’:
main.c:11:24: warning: unused parameter ‘sig’ [-Wunused-parameter]
   11 | void handle_signal(int sig) {
      |                    ~~~~^~~
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-24$ ls -l
```

Interactive Command Processor Execution:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-24$ ./os_processor
Interactive OS Command Processor
Type help for commands. Type exit to quit.
OS> help
Commands: help, pwd, ls, exit
OS> pwd
/home/bhardwaj/2520090182_OSSP/2520090182_Skill/Skill-24
OS> ls
Makefile
hi
os_processor
README.md
main.c
OS> hello
Unknown command: hello
Type help to see available commands.
OS> exit
Exit
```

Linux Pipeline Demonstration:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-24$ ls -l | grep -E 'main.c|README.md|Makefile'
-rw-r--r-- 1 bhardwaj bhardwaj   202 Oct 11 11:47 Makefile
-rw-r--r-- 1 bhardwaj bhardwaj  1061 Oct 11 11:47 README.md
-rw-r--r-- 1 bhardwaj bhardwaj  1902 Oct 11 11:46 main.c
```

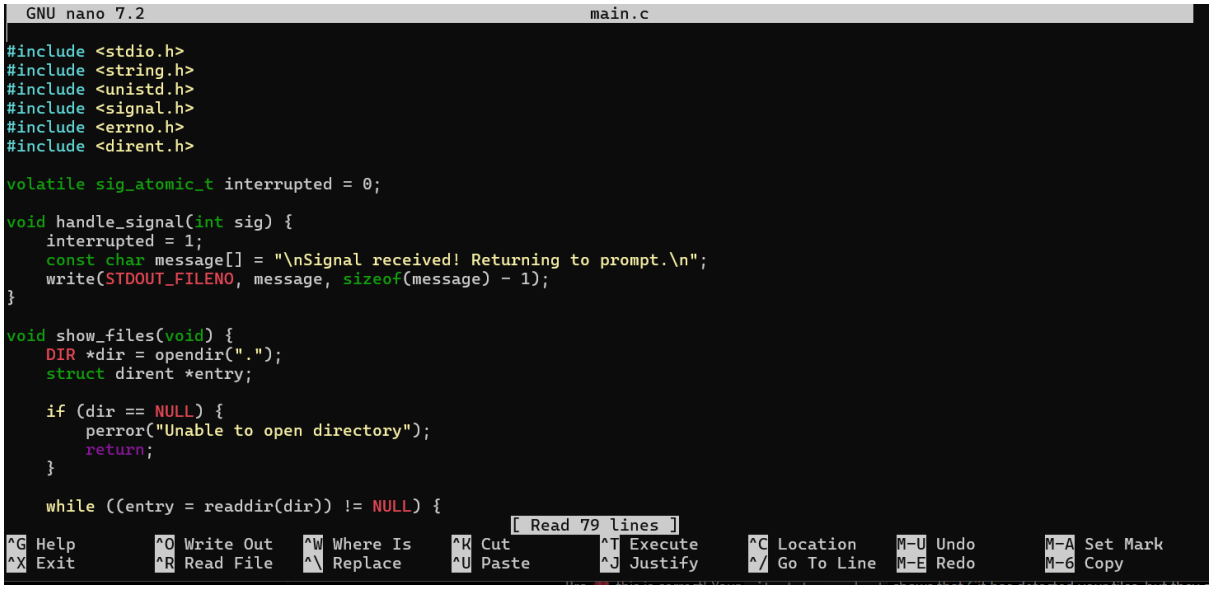
Repository Structure Verification:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-24$ find . -maxdepth 1 -type f -printf '%f\n' | sort
Makefile
README.md
hi
main.c
os_processor
```

Git Commit and Release Tag Verification:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-24$ git status --short
?? Makefile
?? README.md
?? main.c
?? os_processor
```

Code:


GNU nano 7.2 main.c
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <signal.h>
#include <errno.h>
#include <dirent.h>
volatile sig_atomic_t interrupted = 0;
void handle_signal(int sig) {
interrupted = 1;
const char message[] = "\nSignal received! Returning to prompt.\n";
write(STDOUT_FILENO, message, sizeof(message) - 1);
}
void show_files(void) {
DIR *dir = opendir(".");
struct dirent *entry;
if (dir == NULL) {
perror("Unable to open directory");
return;
}
while ((entry = readdir(dir)) != NULL) {
[Read 79 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy